Activity : 1

Match the following

a) Myocardial infarction	-	disruption of blood flow to the brain
b) Plaque	-	chest pain
c) Angina pectoris	-	inadequate blood supply to tissues
d) Ischemia	-	fatty deposits in arteries
e) Stroke	-	heart attack

11.9 Blood lipids

Fatty substances such as cholesterol and triglycerides found in the blood are termed as 'blood lipids'.

- **Cholesterol:** It is a steroid alcohol found exclusively in animals. Our body can synthesize cholesterol in the liver irrespective of whether it is provided in the diet. This is called **"endogenous cholesterol".** A cholesterol level of less than 200mg/dl of blood is desirable (National Cholesterol Education Program (NCEP), 2002).
- **Triglycerides (TG):** Triglycerides are made up of glycerol and fatty acids. They are the main constituents of natural fats and oils. TG is the most common type of fat in the body. A TG level of less than 150 mg/dl of blood is considered desirable (NCEP), 2002). For improved metabolic health and protection to the heart and blood vessels in individuals with increased risk of CVD, the American Heart Association now recommends an optimum fasting TG level of 100 mg/dL (AHA, 2011). AHA also recommends that this may be achieved with diet, weight loss, and increased physical activity.
- **Phospholipids:** It is a lipid containing a phosphate group in its structure. They are an important component of all cell membranes.

11.10 Blood lipoproteins

Lipids, such as cholesterol and triglyceride are insoluble in plasma which is the liquid component of blood. Therefore they bind with circulating lipoproteins that transport lipids to various tissues. Lipoproteins in blood are the mode of transport for cholesterol through the bloodstream and lymphatic fluid. Lipoproteins consist of cholesterol, triglycerides, phospholipids, and protein.

There are five major lipoproteins in the blood

1. **Chylomicrons:** They are the largest and the least dense lipoproteins. They transport triglycerides from the intestine to the cells of the body.
2. **Very low-density lipoprotein (VLDL):** They are responsible for transporting lipids from the liver to the cells of the body.
3. **Intermediate-density lipoprotein (IDL):** This is formed during the conversion of VLDL to LDL.
4. **Low-density lipoprotein (LDL):** These are formed from VLDLs and carry cholesterol from the liver to different cells. They are termed as **'bad cholesterol'** since they aid in the fatty build up within arteries (atherosclerosis) blocking the free flow of blood. High levels of LDL is associated with an increased risk of CVD. LDL cholesterol level below 100 mg/dl of blood is considered ideal. When the level increases beyond this, it increases the risk of heart disease (NCEP, 2002).
5. **High-density lipoprotein (HDL:** They are involved in **"reverse cholesterol transport"** where HDL carries LDL (bad cholesterol) from the arteries to the liver, where it is broken down and eliminated from the body. Therefore,

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